Advanced Optical Fiber Communications

Problem 12: dispersed gaussian pulses

a) A 0.88 μm communication system transmits data over a 10 km single-mode fiber by using 10 ns (FWHM) Gaussian pulses. Determine the maximum bit rate if the light-emitting diode (LED) has a spectral FWHM of 30 nm with Gaussian shape. Use D = −80 ps/(km nm).
b) Estimate the limiting bit rate for a 60 km single-mode fiber link at 1.3 and 1.55 μm wavelength assuming transform-limited 50 ps (FWHM) Gaussian input pulses. Assume $\beta_2 = 0$ and $-20\ ps^2/km$
c) A 1.55 μm optical communication system operating at 5 Gbit/s is using Gaussian pulses of width 100 ps (FWHM) chirped such that C = −6. What is the dispersion-limited maximum fiber length? How much will it change if the pulses are unchirped? Neglect laser linewidth and assume $\beta_2 = -20\ ps^2$ /km.